$a, ar, ar^2, \ldots, ar^{n-1}, \ldots$ where a is the first term and 'r' is the common ratio. Let t_n be the nth term of the G.P.

Then
$$t_1 = a = a \times r^0 = a \times r^{1-1}$$
$$t_2 = t_1 \times r = a \times r = a \times r^{2-1}$$
$$t_3 = t_2 \times r = ar \times r = ar^2 = ar^{3-1}$$
$$\vdots \qquad \vdots$$
$$t_n = t_{n-1} \times r = ar^{n-2} \times r = ar^{n-2+1} = ar^{n-1}$$

Thus, the general term or n^{th} term of a G.P. is $t_n = ar^{n-1}$

Note

If we consider the ratio of successive terms of the G.P. then we have

$$\frac{t_2}{t_1} = \frac{ar}{a} = r\,,\ \frac{t_3}{t_2} = \frac{ar^2}{ar} = r\,,\ \frac{t_4}{t_3} = \frac{ar^3}{ar^2} = r\,,\ \frac{t_5}{t_4} = \frac{ar^4}{ar^3} = r\,,\ldots$$

Thus, the ratio between any two consecutive terms of the Geometric Progression is always constant and that constant is the common ratio of the given Progression.

Progress Check

1. A G.P. is obtained by multiplying _____ to the preceding term.
2. The ratio between any two consecutive terms of the G.P. is _____ and it is called _____.
3. Fill in the blanks if the following are in G.P.

(i) $\frac{1}{8}, \frac{3}{4}, \frac{9}{2}, _____$ (ii) $7, \frac{7}{2}, _____$ (iii) $_____, 2\sqrt{2}, 4, \ldots$

Example 2.40 Which of the following sequences form a Geometric Progression?

(i) 7, 14, 21, 28, … (ii) $\frac{1}{2}$, 1, 2, 4, ... (iii) 5, 25, 50, 75, …

Solution To check if a given sequence form a G.P. we have to see if the ratio between successive terms are equal.

(i) 7, 14, 21, 28, …

$$\frac{t_2}{t_1} = \frac{14}{7} = 2; \qquad \frac{t_3}{t_2} = \frac{21}{14} = \frac{3}{2}; \qquad \frac{t_4}{t_3} = \frac{28}{21} = \frac{4}{3}$$

Since the ratios between successive terms are not equal, the sequence 7, 14, 21, 28, … is not a Geometric Progression.

(ii) $\frac{1}{2}$, 1, 2, 4, ...

$$\frac{t_2}{t_1} = \frac{1}{\frac{1}{2}} = 2; \qquad \frac{t_3}{t_2} = \frac{2}{1} = 2; \qquad \frac{t_4}{t_3} = \frac{4}{2} = 2$$

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